

Lab Assignment No.	5
Title	Develop a MapReduce program to analyze Titanic ship data and to find the average age of the people (only male) who died in the tragedy. How many persons are dead in each class (only female).
Roll No.	
Class	BE
Date of Completion	
Subject	Computer Laboratory-IV : Big Data Analytics
Assessment Marks	
Assessor's Sign	

AIM: Develop a MapReduce program to analyze Titanic ship data and to find the average age of the people (both male and female) who died in the tragedy. How many persons are survived in each class.

The titanic data will be..

Column 1 : PassengerId	Column 2 : Survived (survived=0 & died=1)
Column 3 : Pclass	Column 4 : Name
Column 5 : Sex	Column 6 : Age
Column 7 : SibSp	Column 8 : Parch
Column 9 : Ticket	Column 10 : Fare
Column 11 : Cabin	Column 12 : Embarked

Description:

There have been huge disasters in the history of Map reduce, but the magnitude of the Titanic's disaster ranks as high as the depth it sank too. So much so that subsequent disasters have always been described as "titanic in proportion" – implying huge losses.

Anyone who has read about the Titanic, know that a perfect combination of natural events and human errors led to the sinking of the Titanic on its fateful maiden journey from Southampton to New York on April 14, 1912.

There have been several questions put forward to understand the cause/s of the tragedy – foremost among them is: What made it sink and even more intriguing How can a 46,000 ton ship sink to the depth of 13,000 feet in a matter of 3 hours? This is a mind boggling question indeed!

There have been as many inquiries as there have been questions raised and equally that many types of analysis methods applied to arrive at conclusions. But this blog is not about analyzing why or what made the Titanic sink – it is about analyzing the data that is present about the Titanic publicly.

It actually uses Hadoop MapReduce to analyze and arrive at:

- The average age of the people (both male and female) who died in the tragedy using Hadoop MapReduce.
- How many persons survived – traveling class wise.

This blog is about analyzing the data of Titanic. This total analysis is performed in Hadoop MapReduce.

This Titanic data is publically available and the Titanic data set is described below under the heading Data Set Description.

Using that dataset we will perform some Analysis and will draw out some insights like finding the average age of male and females died in Titanic, Number of males and females died in each

compartment.**DATA SET DESCRIPTION**

Column 1 : PassengerI

Column 2 : Survived (survived=0 & died=1) Column 3 : Pclass

Column 4 : Name

Column 5 : Sex

Column 6 : Age

Column 7 : SibSp

Column 8 : Parch

Column 9 : Ticket

Column 10 : Fare

Column 11 : Cabin

Column 12 : Embarked

Mapper code:

```
public class Average_age {  
    public static class Map extends Mapper<LongWritable, Text, Text, IntWritable> {  
  
        private Text gender = new Text();  
  
        private IntWritable age = new IntWritable();  
  
        public void map(LongWritable key, Text value, Context context ) throws IOException,  
            InterruptedException {  
  
            String line = value.toString();  
  
            String str[]=line.split(",");  
  
            if(str.length>6){  
  
                gender.set(str[4]);  
  
                if((str[1].equals("0")) ){  
  
                    if(str[5].matches("\\d+")){  
  
                        int i=Integer.parseInt(str[5]);  
  
                        age.set(i);  
                    }  
                } }  
            context.write(gender, age)
```

```
}
}
```

Reducer Code:

```
public static class Reduce extends Reducer<Text,IntWritable, Text, IntWritable> {
public void reduce(Text key, Iterable<IntWritable> values, Context context) throws
IOException, InterruptedException { int sum = 0; int l=0; for (IntWritable val : values)
{ l+=1; sum += val.get();
} sum=sum/l; context.write(key, new
IntWritable(sum));
}
}
```

Configuration Code: job.setMapOutputKeyClass(Text.class);

job.setMapOutputValueClass(IntWritable.class);

<https://github.com/kiran0541/Map-Reduce/blob/master/Average%20age%20of%20male%20and%20female%20people%20died%20in%20titanic>

Way to to execute the Jar file to get the result of the first problem statement: *hadoop jar average.jar /TitanicData.txt /avg_out*

Here ‘hadoop’ specifies we are running a Hadoop command and jar specifies which type of application we are running and average.jar is the jar file which we have created which consists the above source code and the path of the Input file name in our case it is TitanicData.txt and the output file where to store the output here we have given it as avg out.

Way to view the output:

hadoop dfs -cat /avg_out/part-r-000000

Here ‘hadoop’ specifies that we are running a Hadoop command and ‘dfs’ specifies that we are performing an operation related to Hadoop Distributed File System and ‘- cat’ is used to view the contents of a file and ‘avg_out/part-r-000000’ is the file where the output is stored. Part file is created by default by the TextInputFormat class of Hadoop.